

Notes on Bioluminescence

1. How the light is made

Bioluminescence is light produced by a chemical reaction inside a living organism. In fireflies the reaction **joins a molecule called luciferin with oxygen**, and an enzyme named luciferase speeds it up. Almost all of the energy leaves as visible light rather than heat, which is why the glow is often **called cold light**. A typical incandescent bulb, by contrast, turns most of its energy into heat.

2. Why fireflies flash

Each firefly species has its own flash pattern, and males use it to signal to females in the grass below. The female answers with a timed flash of her own, so **the delay between signals carries as much meaning as the light itself**. Some predatory fireflies copy the reply of another species to lure males in and eat them. Researchers call these mimics **femme fatale fireflies**.

3. Light in the deep sea

Below about two hundred metres, sunlight fades and most animals make their own light instead. Estimates suggest that roughly **three quarters of deep-sea animals** can produce light in some form. Many fish use **counter-illumination**, glowing faintly on their bellies **so their outline disappears against the dim water above**. The anglerfish hosts glowing bacteria in a lure that dangles in front of its mouth.

4. Uses in the lab

Scientists borrowed the firefly enzyme to build reporter genes that light up when a gene is switched on. **Green fluorescent protein from a jellyfish** became one of the most widely used tools in cell biology. Its discovery and development earned the Nobel Prize in Chemistry in 2008.